

This unit focused on convolutional neural networks (CNNs) and their application in image recognition tasks. While the technical side shows how powerful these models are at identifying patterns in images, it also raises important real-world questions. A good example is the use of facial recognition technology by police forces in the UK, which has been widely discussed in the media.

Reports from the BBC and The Guardian highlighted that these systems were not always accurate and sometimes misidentified individuals. In some cases, people were incorrectly flagged as suspects, which can have serious consequences. There were also concerns about bias, as facial recognition systems have been shown to perform less reliably on certain demographic groups.

What makes this particularly interesting is that the underlying technology is very similar to what we study in CNNs. These models are extremely good at detecting patterns, but they do not “understand” images in a human sense. They learn from data, and if that data contains biases or is not representative, the model will reflect those issues.

Another important aspect is privacy. Facial recognition systems can analyse people in public spaces without their consent, which raises questions about surveillance and civil liberties. The discussions around these systems show that technical performance alone is not enough; how and where these models are used also matters.

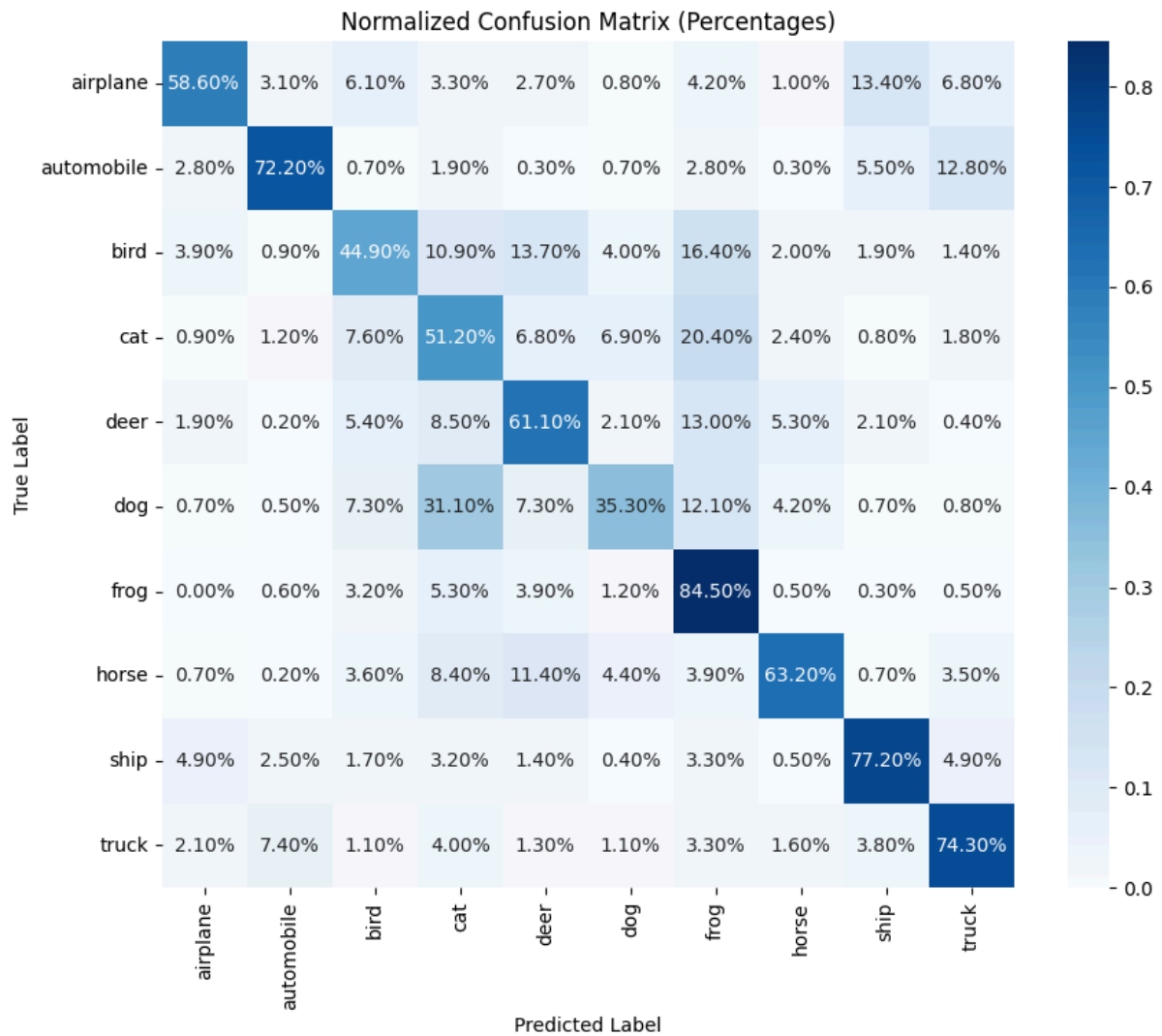
CNN - Object Recognition - Code

The Convolutional Neural Networks (CNN) - Object Recognition.ipynb notebook demonstrates the use of Convolutional Neural Networks (CNNs) for image classification using the CIFAR-10 dataset, which contains 60,000 32×32 colour images across 10 classes. The data is first preprocessed by normalising pixel values and encoding labels into categorical format.

The dataset is then split into training, validation, and test sets. A sequential CNN model is built using convolutional and max pooling layers for feature extraction, followed by a dense layer and a softmax output layer for classification. The model is trained using the Adam optimiser and categorical cross-entropy loss, with EarlyStopping applied to prevent overfitting.

Model performance is evaluated using accuracy and loss curves, as well as a classification report and confusion matrix. Finally, the notebook demonstrates how to make predictions on unseen images.

Based on this notebook's results I have created another confusion matrix. It is like a heat map and now we can see what are the most likely mistakes that the model makes.



Apparently, it is the differentiation of cats and dogs. However, the model is very good at distinguishing between an airplane and a frog. Specifically, the predictions for the first 3 images (in `x_test[ ]`) are correct.